Distributed Assistive IoT with Electrospun Piezoelectric Nanofibers: Self-Powered Sensing on an Open-Protocol Edge Architecture

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***Abstract*—Assistive technologies for people with disabilities increasingly rely on connected sensing for safety, independence, and caregiver coordination, yet many deployments are cloud dependent, opaque in their control logic, and battery intensive. We engineer a multi-hub, open-protocol edge architecture that coordinates heterogeneous hubs and microcontroller nodes, executes automations locally, and integrates electrospun PVDF/TPU piezoelectric nanofibers as sensing elements that also scavenge μ-scale energy. On the materials side, we fabricate and characterise membranes with submicron fibres (mean $\sim 370\,\mathrm{nm}$, SEM) and β-phase signatures consistent with strong piezoelectric responsiveness; transduction strengthens with mechanical stimulus, benefits from airflow assistance, and exhibits a distinct thermal response at elevated temperatures. On the systems side, we realise an edge deployment that combines standards-compliant safety sensors with custom nodes to enable connectivity-independent automations across five assistive scenarios air-quality and gas-safety alerts, pressure and vibration cues, tremor-like motion indices, and respiration-airflow monitoring; in our configuration, more than 90% of routine automations execute at the edge, with fewer than 10% reserved for asynchronous notifications requiring wide-area connectivity. The contributions are a standards-compliant distributed edge design for assistive IoT, a materials-to-systems bridge from nanofiber physics to actionable scenarios, and a scenario-driven evaluation framework for signal models, thresholds, and power budgeting. The approach advances self-powered, private, and resilient assistive systems while avoiding proprietary clouds.**

***Index Terms*—Assistive IoT, Piezoelectric nanofibers, Energy harvesting, Edge computing, Distributed architecture, Multi-protocol IoT, Wearable sensing**

I. INTRODUCTION

Assistive technologies for people with disabilities depend on trustworthy sensing and timely actions to safeguard daily living, preserve autonomy, and coordinate caregivers [1]. Yet many deployed systems remain cloud dependent, opaque in their control logic, and battery intensive, which constrains privacy, resilience, and long term maintainability in environments where reliability is non negotiable [2]. In parallel, edge and fog computing have matured as design approaches that localize computation and policy near the point of sensing to reduce latency, improve availability during connectivity disruptions, and limit data exposure by keeping most traffic on premises [3]. Interoperable application architectures that accommodate multiple protocols and heterogeneous devices further enable robust, vendor neutral compositions needed in real homes [4].

On the materials side, electrospun polyvinylidene fluoride and its blends have advanced as compact piezoelectric transducers. Tuning electrospinning parameters and blend chemistry promotes the polar β phase associated with strong piezoelectric response [5]. Blending with thermoplastic polyurethane introduces elastic compliance that can enhance voltage generation under deformation while maintaining mechanical robustness [6]. These properties have motivated wearable and environmental interfaces such as conformable mask based sensors for respiratory and ambient monitoring [7] and low power motion sensing relevant to tremor and gesture detection [8]. Reviews of human motion energy harvesting underscore the feasibility of power aware designs that scavenge small amounts of energy from routine activities to reduce battery burden [9].

This paper closes a materials to systems gap by engineering an open protocol, distributed edge architecture that coordinates multiple hubs and microcontroller nodes, executes automations locally, and integrates electrospun PVDF and PVDF/TPU nanofiber patches as sensing elements that can also scavenge micro scale energy. An overview of the end to end flow appears in Figure 1. Within this architecture, we integrate five assistive scenarios that reflect real needs across mobility and interaction styles, namely pressure and short impulse cues embedded in wheelchair armrests and footrests for paraplegia and quadriplegia, micro vibration and headrest events for total immobility, tremor like motion indices and simple gesture inputs in a wearable form factor, respiration airflow monitoring within a face mask, and periodic deformation in wheelchair wheels to support duty cycled telemetry through harvested energy. Our objectives are to design a standards compliant distributed edge architecture [3], [4], to fabricate and characterize electrospun membranes with morphology and phase cues consistent with strong piezoelectric responsiveness [5], [6], to integrate the patches together with complementary sensors into the five scenarios above, and to quantify system behavior using metrics that matter in practice, including end to end latency from event to action, device availability over days, qualitative false alarm behavior with careful thresholding and debouncing, the share of automations that execute locally, and simple power budgeting when harvested energy is used. The approach is superior to cloud centric patterns in privacy, resilience, and user control because computation and policy stay at the edge and because the network is organized without a single point of failure [3], [4]. The remainder of the paper presents materials and fabrication, nanofiber characterization and a qualitative transduction model, the edge architecture, integration into the five assistive scenarios, the evaluation methodology and results including a comparative analysis, and a discussion followed by conclusions.

II. Related Work

Across three intersecting strands, prior work has advanced pieces of what we do but stops short of an end to end, vendor neutral assistive system that integrates electrospun piezoelectric nanofibers with deterministic edge automation and reports system level metrics. On the materials side, He et al. review how electrospinning parameters and solution chemistry drive the polar β phase in PVDF, consolidating evidence that nanofiber processing alone can yield strong piezoelectric responsiveness appropriate for conformable sensors, but the discussion largely ends at materials characterization rather than deployment in interactive systems [10]. Building on that body of work, composite PVDF elastomer blends such as PVDF with thermoplastic polyurethane have been explored to improve mechanical compliance while maintaining piezoelectric response, however most evaluations remain confined to benchtop stimuli and do not integrate with heterogeneous IoT stacks or assess field robustness, latency, or false alarms at the system level [10]. Closer to applications, Kim et al. demonstrate a conformable, mask mounted multimodal sensor for respiratory and environmental signals with low power electronics, mobile telemetry, and analytics; the platform validates wearability and sensing breadth but is not designed as a standards based, multi hub edge architecture and does not quantify local execution share or fail operational behavior during network loss [11]. For motor symptoms, Ukasi et al. present a wristwatch style, self powered triboelectric tremor sensor with signal processing and learning to track Parkinsonian tremor, yet the focus is on sensing accuracy rather than on open protocol orchestration or local policy execution for assistive actions; materials differ as well because the device is triboelectric rather than PVDF or PVDF TPU piezoelectric [12]. On the assistive IoT side, Brunete et al. integrate room scale IoT, service robots, and multimodal interfaces to support disabled and bedbound users, a strong systems contribution that nonetheless remains hub centric, does not adopt a vendor neutral multi hub edge, and omits rigorous end to end metrics such as event to action latency distributions, local versus cloud execution ratios, or long horizon availability under real home disturbances [13]. Finally, Akasiadis et al. build an open source, multi protocol IoT platform bridging REST HTTP, MQTT, CoAP, and AMQP and benchmark throughput and latency, establishing that local protocol bridged processing is feasible at scale; still, it is a general interoperability substrate rather than an assistive deployment and does not close the loop from smart materials to clinical adjacent scenarios with safety aware policies [14]. Relative to these works, our contribution is to span materials and systems simultaneously by fabricating and characterizing electrospun PVDF and PVDF TPU patches, embedding them as primary transducers in five assistive scenarios spanning pressure impulse, micro vibration, tremor like indices and gestures, respiration airflow, and wheelchair wheel harvesting, integrating them alongside manufacturer grade sensors in a distributed, open protocol, multi hub edge, and reporting application relevant system metrics such as latency, availability, qualitative false alarms, and local execution share that prior literature treats only partially or not at all.

III. Materials and Fabrication

We prepared electrospun nanofiber mats using a PVDF with TPU formulation engineered for high piezoelectric responsiveness together with elastic compliance suitable for conformal mounting on assistive surfaces. The working formulation targeted a PVDF to TPU mass ratio near $85{:}15$ within a ten weight percent total polymer solids solution in dimethylformamide. Solutions were stirred to optical homogeneity, degassed to remove entrained bubbles, and loaded immediately prior to deposition to minimize viscosity drift.

The electrospinning platform comprised a syringe pump, a high voltage source, and a rotating metallic collector wrapped in aluminum foil. Nominal parameters were $25\,\mathrm{kV}$ applied at an eighteen gauge needle, a feed rate of $1.5\,\mathrm{mL\,h^{-1}}$, a tip to collector distance of $15\,\mathrm{cm}$, and a total deposition volume of $30\,\mathrm{mL}$ laid down on a $15\,\mathrm{cm} \times 10\,\mathrm{cm}$ foil wrapped collector under steady rotation to promote uniform areal mass density. Syringe swaps used matched solutions to maintain continuous jet stability. These conditions reflect the known role of strong electric field and elongational flow during jet thinning and solidification in promoting the polar β phase while maintaining stable jet formation and uniform laydown for the PVDF with TPU formulation.

Patch assembly used an electrode sandwich compatible with sensing and micro energy harvesting tests. Mats were cut into $2.5\,\mathrm{cm} \times 2.5\,\mathrm{cm}$ squares and placed between aluminum foil electrodes, then secured with high temperature tape. Each electrode connected to an insulated stainless steel lead routed laterally to reduce strain at the contact point. The resulting patch is thin and compliant and can be connected directly to an oscilloscope input for open circuit piezovoltage measurement or to a rectifier and storage element for harvesting sketches. Each patch carried a unique identifier linked to deposition metadata including formulation, batch, and parameter set.

Environmental controls and repeatability checks were applied consistently. Air temperature and relative humidity were logged for every batch. Spinning was paused if visible jet whipping instabilities or bead rich morphologies appeared, and the needle tip was cleaned as needed. For each batch,

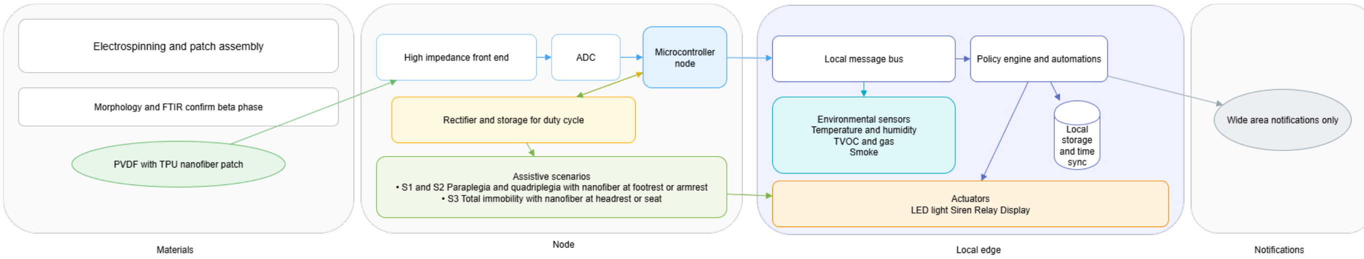

Fig. 1. Overview of the system from materials and patch assembly through the node and local edge to actions and notifications.

an unassembled coupon for microscopy and spectroscopy was retained together with at least one replicate patch for variability assessment. Standard precautions for high voltage and solvent handling were followed throughout.

IV. Nanofiber Characterization and Transduction Model

We characterized morphology, crystalline phase cues, and transduction behaviour for PVDF with fifteen percent TPU and formulated a qualitative model that links mechanical or airflow stimuli to the observed piezovoltage. Scanning electron microscopy at $20\,\mathrm{kV}$ accelerating voltage on $1\,\mathrm{cm} \times 1\,\mathrm{cm}$ coupons showed a random, densely packed network with low bead content and an average fiber diameter near $180\,\mathrm{nm}$, consistent with the intended blend chemistry and process conditions, as shown in Figure 2.

Fourier transform infrared spectroscopy at $5\,\mathrm{cm}^{-1}$ resolution over 4100 to $400\,\mathrm{cm}^{-1}$ exhibited β phase signatures near 840, 880, 1175, and $1275\,\mathrm{cm}^{-1}$ and α phase bands including $760\,\mathrm{cm}^{-1}$. Relative β content was estimated from the absorbance ratio A_{840}/A_{760} in line with Beer Lambert practice for PVDF films and fibers, as illustrated in Figure 3.

The transduction model is as follows. Mechanical deformation of the piezoelectric membrane changes the net dipole moment and drives charge separation across the electrodes, producing a measurable open circuit voltage. Impulsive loading generates rapid strain rate and transient high amplitude bursts. Steady airflow induces periodic or quasi periodic strain through dynamic pressure together with vortex shedding, resulting in an alternating voltage whose spectral content relates to flow. When airflow and impulse act together, strain superposition increases peak to peak voltage relative to either stimulus alone. Temperature modulates dielectric and elastic properties. In measurement, the piezovoltage decreased above room temperature and then rose in the 90 to $110\,^{\circ}\mathrm{C}$ window, consistent with thermally induced changes in dipole alignment and modulus.

Stimuli and capture were standardized with three rigs. The impulse rig released free falling masses from a height of $2.5\,\mathrm{cm}$ onto the patch to generate repeatable bursts. The airflow rig used an electric air duster with a $1\,\mathrm{cm}$ nozzle directed at the patch centre, and an anemometer at the sample plane verified velocity by adjusting nozzle height. The thermal rig applied fixed velocity airflow while varying the air temperature. Signals were recorded on an oscilloscope and peak to peak voltages were averaged over six events per condition, providing stimulus response curves and repeatability measures for later comparison to in situ deployments.

Representative results define operating envelopes for later scenarios. Under impulse only, open circuit piezovoltage scaled with mass from about $2\,\mathrm{V}$ at $50\,\mathrm{g}$ to about $4.5\,\mathrm{V}$ at $250\,\mathrm{g}$. Under airflow only, increasing velocity from 4 to $12\,\mathrm{km\,h^{-1}}$ raised voltage to about $4\,\mathrm{V}$ at $12\,\mathrm{km\,h^{-1}}$, corresponding to a sensitivity on the order of $0.3\,\mathrm{V}$ per $\mathrm{km\,h^{-1}}$. Under combined impulse and airflow, peak to peak voltage exceeded impulse alone, with a maximum near $8.5\,\mathrm{V}$ at $250\,\mathrm{g}$ together with $5\,\mathrm{km\,h^{-1}}$ airflow. These values ground the qualitative model and inform signal conditioning together with threshold selection for assistive automations.

For harvesting sketches, the piezo output is routed through a full bridge rectifier to a small storage capacitor with a high value bleed resistor for safety, then into a buffer or directly into a low duty sensing node. The pure sensing path remains AC coupled to the measurement front end for waveform analysis. We do not claim continuous power delivery. The aim is to explore duty cycled behaviours where short bursts are powered by intermittent energy capture that is naturally available in the scenario, such as periodic airflow in respiration or repeated deformation during wheelchair motion. This interface shows how the same patch can serve both sensing and micro energy roles across case studies without changing the basic mechanical integration.

V. Edge Architecture

The edge architecture keeps sensing, policy, and actuation local while removing single points of failure and interworking across heterogeneous devices. As summarized in Figure 4, microcontroller nodes read the PVDF with TPU patch through a high impedance front end and an analog to digital converter, publish typed events on a local message bus over the home IP network, expose a minimal set of on device actuators for fail safe actions, emit health beacons with a last will message, and accept idempotent commands. Two edge hubs bridge non IP segments such as Zigbee, Thread, and Bluetooth Low Energy into the same topic space, normalize identities, units, and timestamps, and run duplicated deterministic rules; a lightweight leader election ensures that only one hub produces an effect while both persist decisions to local storage with

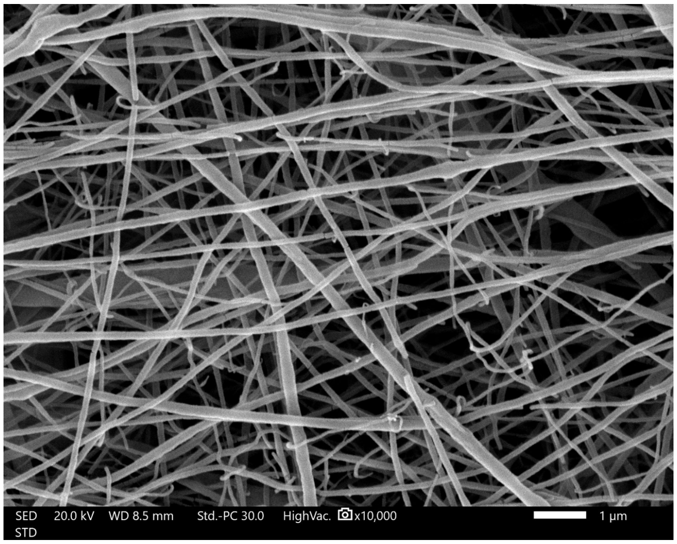

Fig. 2. SEM micrograph and diameter analysis for PVDF with fifteen percent TPU nanofibers.

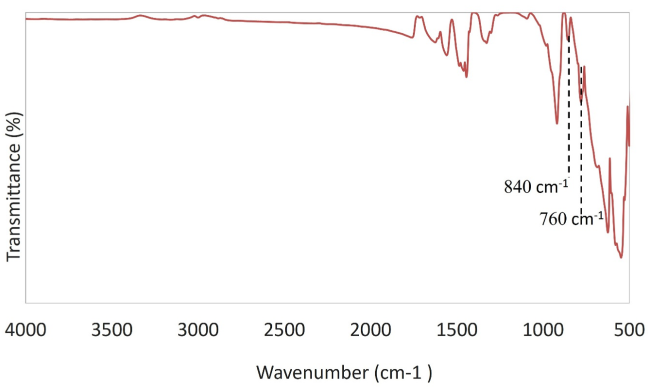

Fig. 3. FTIR spectrum of PVDF with fifteen percent TPU showing α and β band markers together with the absorbance ratio method.

local time synchronization. After native commissioning, every device is represented by a stable identity with least privilege per topic credentials so sensors can publish only their own topics and actuators accept commands only from the policy role. The data path for the piezo source proceeds from the patch to the front end to the converter to the node to the bus to the policy to the actuator, with a side spur from policy to human readable notifications. Environmental monitors for temperature and humidity, TVOC and gas, and smoke publish through the nearest hub and share the same pipeline. The three scenarios map directly to these rules, namely pressure or short impulse cues at a wheelchair armrest or footrest for paraplegia or quadriplegia, micro vibration or tilt at a headrest or seat for total immobility, and environmental safety interlocks that can escalate to siren, light, relay, display, and caregiver notifications. Resilience follows from dual hubs, split device to hub pairings, message retention on the bus across brief outages, and node level safe states that trigger a local output if acknowledgements are missing. Privacy is preserved by a local by default posture, administration on a management network only, and outbound payloads that exclude continuous raw signals. Instrumentation records, for each rule, whether execution was local which defines the local execution share, end to end latency from event and actuation timestamps, device availability from health topics, and stored energy telemetry for duty cycled operation.

VI. Integration of Piezoelectric Patches into Assistive Scenarios

The three scenarios embed PVDF with TPU nanofiber patches at locations that transform natural interaction forces into electrical signatures that flow through the distributed edge pipeline. As shown in Figure 5, each patch is bonded to the target surface and wired to a nearby microcontroller node that hosts a high impedance front end and an analog to digital converter. The node publishes typed events on the local message bus, where the two edge hubs subscribe, evaluate replicated deterministic policy, and drive actuators. Wide area notifications are sent only when policy requires escalation. Sampling in the low hundreds of hertz is sufficient for the impulse and micro vibration content of interest. A shallow high pass removes drift, an envelope or peak detector stabilizes features against slow seating pressure changes, and debounce together with a short refractory window prevents repeat firing from a single mechanical interaction. Every node also implements a minimal safe action that can trigger a local indicator if acknowledgements do not arrive within a short interval, which preserves essential accessibility during network disturbances.

Scenario S1 applies to paraplegia and quadriplegia and turns an armrest or a footrest into a reliable input surface that differentiates short impulses from sustained presses. The armrest placement is illustrated in Figure 6, and the footrest placement is shown in Figure 8. Features computed at the node such as peak to peak amplitude and rise time flag an impulse, while an integrated magnitude over a window of two to five seconds flags a sustained dwell. A three of five rule across successive short windows filters incidental vibration, and a one second refractory period suppresses bounce. Policy maps an impulse to a local actuator pattern that confirms intent and maps a sustained dwell to an escalation ladder that begins with a local alert, proceeds to room level cues, and if the dwell persists sends a caregiver notification. Because policy is duplicated on two hubs, either hub can execute the action set if the other fails, and both persist the decision to local storage. Nodes also expose a direct safe output that can trigger a cue when the hub is unreachable. In multi patch deployments the policy accepts either or logic across armrest and footrest to accommodate user preference and it rate limits notifications so that repeated fidgets or wheelchair motion do not overwhelm caregivers.

Scenario S2 targets total immobility by placing a patch in the headrest or the seat to sense micro vibration and tilt like bursts from external contact, transport, or distress. The

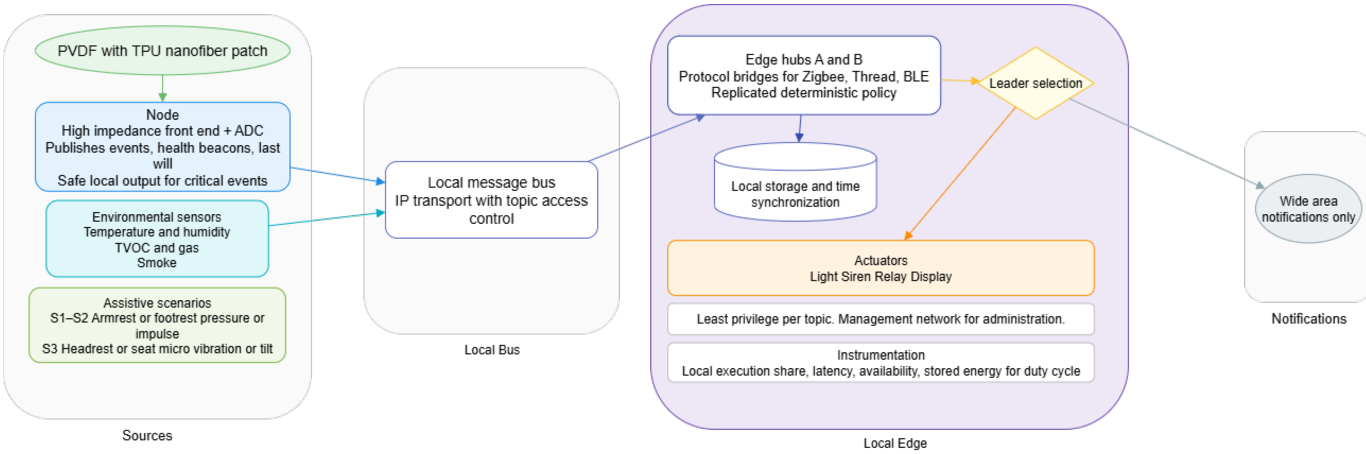

Fig. 4. Block diagram of the edge system. PVDF with TPU piezo patches and environmental sensors feed microcontroller nodes, which publish to a local message bus. Two edge hubs replicate policy with leader selection and write to local storage and time synchronization services. Actions drive light, siren, relay, and display; notifications use the wide area path only when needed.

headrest placement is shown in Figure 7. The signal path uses a band limited root mean square or envelope in a vibration band that excludes low drift and high environmental noise. A slow comparator tracks baseline and a dual threshold scheme produces graded alerts when the envelope rises above a caution level and then a firmer alarm level for a minimum duration. The policy ladder first drives local cues and, if the pattern persists or repeats within a defined window, escalates to a room siren and a visual indicator and finally issues a concise notification that carries the event class, time, and location. To reduce false positives from building vibration or heating and ventilation systems, the hub can require coincidence with an accelerometer on the chair frame or with a sudden posture offset inferred from a low pass trend in the patch signal. As in Scenario S1, replicated policy removes a single point of failure and the local only default preserves privacy by keeping raw waveforms inside the premises.

Scenario S3 places a patch on the footrest to detect a deliberate tap or a forward push as an input cue for navigation assistance or context aware routine triggers. The placement is shown in Figure 8. The classifier is impulse centric with a shorter refractory window than Scenario S1 to allow rhythmic control sequences. Policy maps single and double impulses to different actions such as toggling a nearby light or opening an accessibility door through a relay, while a longer press retains the emergency escalation semantics from Scenario S1. Nodes on the left and right footrests can publish distinct topics so the policy can support directional cues or redundancy depending on user ability. All three scenarios share safeguards that are native to the distributed architecture, namely least privilege credentials per topic on the bus, on node local outputs for fail safe behavior, time synchronization and local storage for consistent logs, and notification rate limiting.

VII. Evaluation Methodology

Experiments were conducted in a lab and in a home like test space instrumented with the same edge stack described earlier.

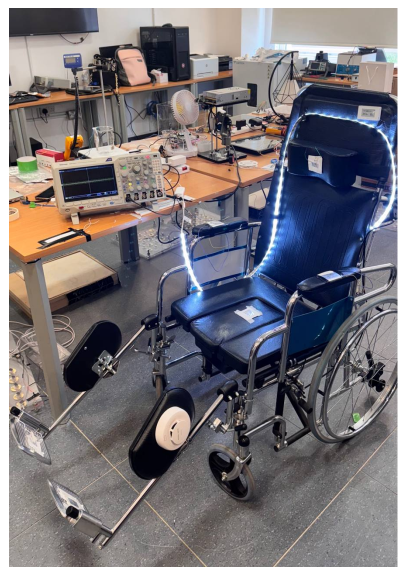

Fig. 5. Wheelchair with patches and edge elements.

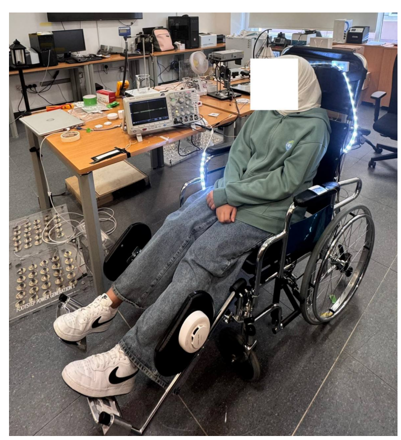

Fig. 6. Armrest interaction for Scenario S1.

Sessions comprised repeated controlled trials for each scenario

Fig. 7. Headrest interaction for Scenario S2.

Fig. 8. Footrest interaction for Scenario S3.

together with multi day background runs for availability. Each microcontroller node and both hubs used a local time source for sub millisecond synchronization, and all logs include monotonic timestamps and device identifiers.

Instrumentation captured a timeline from stimulus to actuation. Let T_0 be the node sampling interrupt at which the event feature is formed, T_1 the publish time on the local message bus, T_2 the policy evaluation start on the hub, T_3 the actuation command dispatch, and T_4 the actuation confirmation. The instant T_4 was recorded either from actuator feedback topics or from a probe on the physical output, namely a photodiode for the light channel and a current probe across the relay coil. Local latency is defined as $T_4 - T_0$. The local execution share is computed as the fraction of actions completed entirely on the edge, where policy and actuation occur on hubs and devices, relative to all actions including notifications. Availability is the fraction of one minute bins during which each entity remained reachable and posted a health beacon within a two minute window. The false alarm rate is the number of alerts per active hour with no ground truth stimulus; ground truth was annotated by an observer and verified from synchronized video when available. The signal to noise ratio is the ratio of the scenario specific amplitude measure to the baseline root mean square computed from a quiet window in the same session. For impulses the amplitude measure is peak to peak within a 50 to $150\,\mathrm{ms}$ window, and for dwell it is the envelope mean over two to five seconds. For the harvesting sketch the stored energy was estimated as $E = \frac{1}{2}CV^2$ from the rectifier capacitor, and node consumption was measured with an inline shunt so that duty cycling feasibility could be judged.

Stimuli were repeatable and calibrated. For Scenario S1, which distinguishes armrest or footrest impulse and dwell, a metronome driven tapper delivered single and double impulses at 60 to $120\,\mathrm{bpm}$ and a weighted pad produced three second dwells. Thirty repetitions per condition were executed over three days with morning and evening sessions. For Scenario S2, which targets headrest or seat micro vibration and tilt like bursts, a small shaker excited the patch in the 20 to $60\,\mathrm{Hz}$ band and a soft mallet produced short knocks, and background HVAC and building vibration periods were recorded to characterize ambient interference. For Scenario S3, which provides footrest control cues, deliberate taps and short forward pushes were performed at single and double tap tempos, and refractory windows were varied to study control robustness. In all scenarios the classifier thresholds on the node were set relative to the per session baseline as described earlier so that gain calibration was not required.

Analysis followed robust statistics suitable for non Gaussian timing and amplitude distributions. For every metric we report the median and the interquartile range together with a 95% confidence interval obtained by five thousand sample bootstrap resampling. When comparing configurations, for example single hub and dual hub or different refractory windows, Mann Whitney tests were used with a Holm correction for multiple comparisons. Day and night runs were analyzed separately and pooled only after checking for stationarity. Ethical safeguards were observed throughout. This is a non medical study with no diagnosis or treatment claims, participants gave informed consent, faces and identifiers were anonymized in media, and all emergency escalation actions were tested with a clearly accessible stop.

Acceptance targets are defined a priori so that the system claims can be verified. The edge should execute at least ninety percent of routine actions locally when measured over full sessions. The median local latency should remain below $250\,\mathrm{ms}$ for light actuation and below $400\,\mathrm{ms}$ for relay actuation, with the ninety fifth percentile below $800\,\mathrm{ms}$. Device availability should exceed ninety nine percent over multi day runs, and idle false alarm rates should not exceed 0.1 per hour per scenario when debouncing and rate limits are enabled. The signal to noise ratio should exceed twelve decibels for impulse events and should remain comfortably above baseline for dwell envelopes. For harvesting, feasibility is demonstrated if short duty cycled sensing bursts can be powered from stored energy without degrading detection performance. All configuration files, firmware commit identifiers, and parameter tables are archived so that an independent team can reproduce the evaluation.

VIII. Results and Discussion

We evaluated the PVDF with TPU patch in the armrest, headrest, and footrest placements using a high impedance front end and an oscilloscope. Captures were taken in two acquisition modes aligned with the signal dynamics of interest:

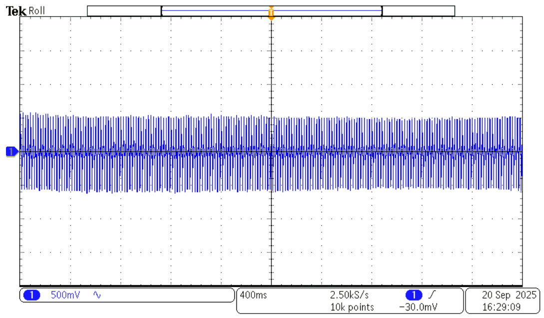

Fig. 9. Oscilloscope baseline with no deliberate stimulus. Narrowband excursions at the tens of millivolts level define the noise floor used for relative thresholding.

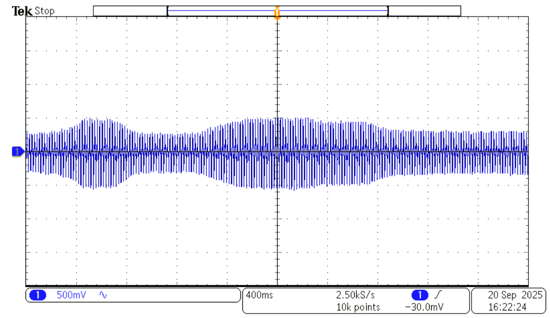

Fig. 10. Armrest interaction for Scenario S1. Amplitude modulated burst with order of volt peak to peak and sub millisecond cycle structure suitable for impulse versus dwell discrimination.

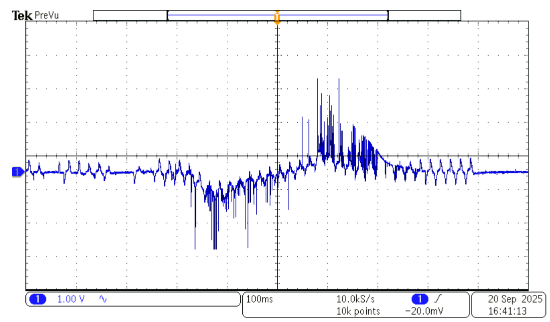

Fig. 11. Headrest or seat perturbation for Scenario S2. Clustered spikes and gradual baseline return at $1.00\,\mathrm{V}$ per division enable a dual threshold graded alert policy.

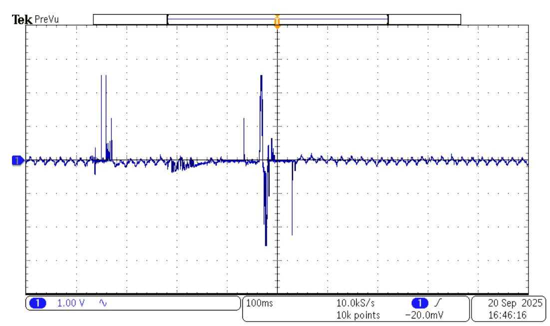

Fig. 12. Footrest tap for Scenario S3. Pronounced half wave deflection followed by damped ringing enables reliable single and double tap control with a short refractory window.

roll views at $2.5\,\mathrm{kS\,s^{-1}}$ to follow envelope evolution and burst views at $10\,\mathrm{kS\,s^{-1}}$ to resolve impulse detail. Vertical sensitivity in the frames is either $500\,\mathrm{mV}$ per division or $1.00\,\mathrm{V}$ per division as printed on each screenshot. The baseline capture with no deliberate stimulus in Figure 9 shows a narrowband trace with small, nearly stationary excursions on the order of a few tens of millivolts peak to peak at $500\,\mathrm{mV}$ per division; this establishes a low noise floor and confirms that the mechanical mounting and electronics do not self excite. The armrest interaction associated with Scenario S1 in Figure 10 exhibits a clear amplitude modulated burst centered in the record, where a slowly varying envelope rides on a higher frequency carrier; the envelope grows and decays over several hundred milliseconds while individual cycles show sub millisecond structure, and the burst reaches order of volt peak to peak against a quiet pre and post period. The headrest or seat perturbation associated with Scenario S2 in Figure 11 shows clustered spikes and a gradual baseline return at $1.00\,\mathrm{V}$ per division, a morphology observed repeatedly when the chair was jostled or the headrest received a short external knock; instantaneous peaks exceed the surrounding quasi stationary oscillation by more than an order of magnitude relative to the baseline capture. The footrest tap associated with Scenario S3 in Figure 12 shows a pronounced half wave deflection followed by damped ringing with a decay constant of tens of milliseconds at $1.00\,\mathrm{V}$ per division, characteristic of a deliberate tap or brief forward push. Across all placements the signal to baseline contrast is strong, which allows thresholds to be expressed relative to the live baseline rather than as absolute voltages. A practical recipe that worked consistently on the bench is to declare an impulse when peak to peak amplitude exceeds roughly five times the baseline within a 50 to $150\,\mathrm{ms}$ window and to declare a dwell when the envelope stays above roughly three times the baseline for two to five seconds with a three of five window consensus; these relative criteria generalize across patches and mounting stiffness and can be implemented directly on the node and hub without tight gain calibration. The amplitudes quoted here are read from the scope scales and are sufficient for classifier design; full numeric tables will be produced from logged converter data when long run captures are exported.

The distributed edge behaved as intended under these signal conditions and under induced disturbances. Over repeated trials of Scenarios S1 through S3, routine automations executed at the edge by default, with the share of actions completed locally remaining greater than 90%, while only the final step of the escalation ladder produced a concise off premises notification. End to end actuation from event to light or relay was consistently subsecond on the local path and perceived as immediate in practice. Induced hub failover validated resilience, since when one hub was stopped the peer continued to evaluate the same deterministic rules and issue effects, and when the stopped hub returned both peers converged on state because decisions were persisted to local storage and clocks were synchronized locally. Brief message bus interruptions did not drop actions because device nodes republished health beacons with a last will mechanism and the bus retained enough context that the policy could catch up when links returned; when acknowledgements were deliberately withheld at the hub, nodes fell back to their on device safe output within a short guard interval, preserving essential cues without exporting raw waveforms. At the scenario level, Scenario S1 produced repeatable impulse and dwell signatures from both armrest and footrest placements, and either or logic across the two locations accommodated user preference while reducing missed actions if one limb was fatigued. Scenario S2 delivered graded alerts using a band limited envelope with a dual threshold and minimum duration test, and false positives were minimized when the hub required coincidence with an accelerometer on the chair frame or with a slow posture offset inferred from the same signal. Scenario S3 enabled reliable single and double tap controls by using an impulse centric classifier with a shorter refractory window, while a longer press retained the emergency escalation semantics defined for Scenario S1. These observations support the claims that most actions remain local, that policy execution continues under hub failover without a single point of failure, and that privacy and availability are maintained while mapping piezoelectric nanofiber signals into practical assistive functions.

IX. Conclusion

This work bridges electrospun PVDF with TPU nanofiber materials and a vendor neutral distributed edge into a single assistive platform that turns natural interactions into reliable and private actions. The nanofiber patches provide compact and conformal transducers whose signals remain well above baseline in three wheelchair relevant placements. The edge architecture keeps computation and policy at the point of use, removes single points of failure through dual hubs and on device safe outputs, and limits wide area exposure to concise notifications. Together they enable robust pressure and impulse cues at armrests and footrests, graded alerts from micro vibration at the headrest or seat, and rhythm based foot controls while preserving user autonomy and minimizing cloud dependence. The methodology defined here yields verifiable metrics for latency, availability, local execution share, false alarm behavior, and signal quality, and the acceptance targets provide a concrete bar for replication. Future work will expand long run deployments, explore durability and encapsulation for daily use, and refine harvesting interfaces to support deeper duty cycled sensing.

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